

INVITED TALK / ABSTRACT

Isoparametric functions and Hardy-Sobolev type equations on riemannian manifolds

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Abstract

Let (M, g) be a closed Riemannian manifold of dimension at least 3. Let S be the union of the focal submanifolds of an isoparametric function on (M, g) . In this talk we are going to discuss results on the existence of solutions of the Hardy-Sobolev type equation

$$\Delta_g u + K(x)u = \frac{u^{q-1}}{(d_S(x))^s},$$

where $d_S(x)$ is the distance from x to S , K is a continuous function on M and $q > 2$. In particular, we will prove the existence of infinite sign-changing solutions to this equation. This talk is based on joint work with Jimmy Petean (CIMAT, Guanajuato, Mexico).

Keywords: isoparametric functions; Hardy-Sobolev type equations; Riemannian manifolds; focal submanifolds; existence of solutions; infinite sign-changing solutions